CompTIA Security+ Exam SY0-701

Lesson 15

Explain Risk Management Processes

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Topic 15A

Risk Management Processes and Concepts

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Risk Identification and Assessment

- Risk identification is fundamental to managing risk
  - Malware attacks
  - Phishing attempts
  - Insider threats
  - Equipment failures
  - Software vulnerabilities
  - Nontechnical risks like inadequate policies or training
- Risk assessment evaluates previously identified risks to determine their potential impact on the organization

Risk Identification and Assessment

- Risk Analysis
  - Describes identifying and evaluating potential risks and the characteristics that define them
- Quantitative Analysis
  - Assign tangible values to each risk
- Qualitative Analysis
  - Assess risks based on subjective judgment
- Risk Assessment
  - Estimates potential risk levels and their significance by interpreting data collected during risk analysis

Risk Identification and Assessment

Risk Factor	Impact	ARO	Cost of Controls	Overall Risk
Legacy Windows Clients	⚠	✗	⚠	✗
Untrained Staff	✓	⚠	✓	⚠
No Antivirus Software	⚠	✗	⚠	✗

Traffic light impact grid.

- Inherent Risk
  - Level of risk before any type of mitigation has been attempted
- Heat Map

Risk Management Strategies

- Risk management strategies
  - Describe the proactive and systematic approaches used to identify, assess, prioritize, and mitigate risks to minimize their negative impacts
- Risk responses
  - Identify how risk items are managed

The four risk responses are avoid, accept, mitigate, and transfer.

Risk Management Strategies

- Residual Risk
  - The amount of risk left after mitigations are implemented
  - Risk cannot be fully eliminated
- Risk Appetite
  - Acceptable levels of risk
  - Varies from one organization to another
  - Sometimes defined in a formal risk appetite statement

Risk Management Processes

- Identifying, assessing, and mitigating vulnerabilities and threats to the essential functions that a business must perform to fulfill its purpose
- Risk
  - A measure of threats, vulnerabilities, impact, and probability
- Risk Registers
  - Risk description
  - Severity
  - Owner of the risk item
  - Identified mitigations
  - Often utilize heat maps

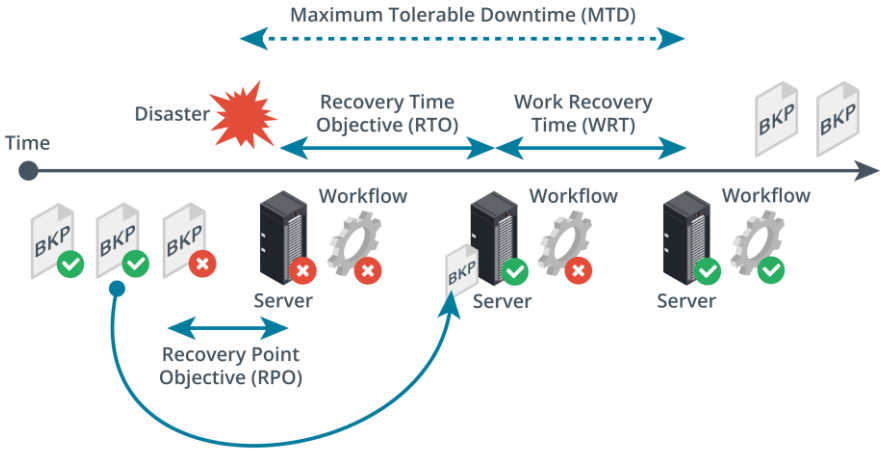
Risk Management Processes

- Risk Threshold
  - Defines the limits or levels of acceptable risk
- Key Risk Indicators
  - Predictive indicators for monitoring and predicting potential risks
- Risk Reporting
  - Communicate an organization's risk profile
  - Communicate the effectiveness of a risk management program

Business Impact Analysis

- Identification of Critical Systems
- Mission Essential Functions

- Maximum Tolerable Downtime (MTD)
- Recovery Time Objective (RTO)
- Work Recovery Time (WRT)
- Recovery Point Objective (RPO)



Metrics governing mission essential functions. (Images © 123RF.com.)

Review Activity: Risk Management Processes and Concepts

- Risk Identification and Assessment
- Risk Management Strategies
- Risk Management Processes
- Business Impact Analysis

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Topic 15B

Vendor Management Concepts

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Vendor Selection

- Systematically evaluate and assess potential vendors to minimize risks associated with outsourcing or procurement
- Third-Party Vendor Assessment
  - Critical component of Governance, Risk, and Compliance (GRC)
  - Vendor assessments provide evidence of due diligence
- Conflict of Interest
  - When an individual or organization has competing interests or obligations that could compromise their ability to act objectively, impartially, or in the best interest of the organization

Vendor Selection

- Vendor Assessment Methods
  - Evidence of Internal Audits
  - Independent Assessments
  - Penetration Testing
  - Supply Chain Analysis
  - Right-to-Audit Clause
- Vendor Monitoring
  - Continuously evaluating vendors to ensure ongoing adherence to security standards, compliance requirements, and contractual obligations

Legal Agreements

- Initial Agreements
  - Memorandum of Understanding (MOU)
  - Nondisclosure Agreement (NDA)
  - Memorandum of Agreement (MOA)
  - Business Partnership Agreement (BPA)
  - Master Service Agreement (MSA)
- Operational/Performance Agreements
  - Service-level Agreement (SLA)
  - Statement of Work (SOW)/Work Order (WO)
- Expectations
  - Rules of Engagement (RoE)

Review Activity: Vendor Management Concepts

- Vendor Selection
- Vendor Assessment Methods
- Legal Agreements

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Topic 15C

Audits and Assessments

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Attestation and Assessments

- Attestation
  - Verifying the accuracy, reliability, and effectiveness of security controls
- Internal Assessment
  - Organization's own employees conduct an in-depth assessment
  - Relatively simple to perform and customize
- External Assessment
  - Independent Third-Party
  - Impartial and objective evaluation of business practices
  - Required for legal compliance

Penetration Testing

- Uses authorized hacking techniques to discover exploitable weaknesses in the target's security systems.
- Sometimes referred to as Pen Test or Ethical Hacking
- Internal Pen Test performed by a “Red Team”
- May include Active and Passive Reconnaissance
- Known Environment Penetration Testing
- Partially Known Environment Penetration Testing
- Unknown Environment Penetration Testing

Exercise Types

- Different types of penetration tests allow organizations to use a flexible and prioritized approach toward security assessment
- Offensive Penetration Testing “Red Team”
- Defensive Penetration Testing “Blue Team”
- Physical Penetration Testing
- Integrated Penetration Testing
  - Combines different types of penetration testing techniques

Review Activity: Penetration Testing Concepts

- Attestation and Assessments
- Penetration Testing
- Exercise Types

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Summary

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